

USER DOCUMENTATION

SOW Analysis Tool

Automated Statement of Work and Bill of Quantities analysis powered by AI. Generate WBS, checklists, questionnaires, deployment plans, and risk assessments from any SOW/BOQ document in minutes.



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01 What is the SOW Analysis Tool?

An interactive command-line tool that reads your Statement of Work (SOW) or Bill of Quantities (BOQ) document and uses AI to automatically generate project planning artifacts - saving hours of manual work.

Document Parsing

Reads PDF, DOCX (SOW) and XLSX (BOQ) files. Extracts text, sections, and requirements automatically.

AI-Powered Analysis

Uses Claude (AWS Bedrock) to analyze services, architecture, cost implications, and risks from the document.

Multi-Output Generation

Generates WBS, Checklist, Questionnaire, Deployment Plan, Risk Analysis, and SOW Analysis in one run.

Interactive CLI

Beautiful terminal interface with progress spinners, color output, and step-by-step guided workflow.

RAG Knowledge Base

Automatically syncs parsed chunks to LanceDB for future retrieval and cross-project learning.

One-Click Install

Single command installation for Windows (PowerShell) and Linux/Mac. No manual setup required.

Who is this for? Implementation engineers, project managers, pre-sales architects, and delivery leads who need to quickly extract actionable plans from SOW/BOQ documents before project kickoff.

02 System Requirements

The tool runs on Windows, Linux, and macOS. It requires Node.js for the interactive CLI and Python for the AI pipeline.

COMPONENT	MINIMUM	RECOMMENDED	NOTES
Operating System	Windows 10, Ubuntu 20.04, macOS 12	Windows 11, Ubuntu 22.04+, macOS 14+	Any x86_64 or ARM64 system
Node.js	v18.0.0	v22 LTS	Required for the interactive CLI
Python	3.10	3.12	Required for the AI pipeline
RAM	4 GB	8 GB	More RAM helps with large docs
Disk Space	500 MB	1 GB	Tool + packages + outputs
Internet	Required	Stable broadband	For AWS Bedrock API calls

Python Packages (auto-installed)

PACKAGE	PURPOSE
boto3	AWS SDK - communicates with Bedrock AI models
openpyxl	Read/write Excel files (BOQ input, XLSX output)
markdown	Convert PDF/DOCX to structured markdown
tiktoken	Token counting for AI prompt management

PACKAGE	PURPOSE
<code>python-dotenv</code>	Environment variable management
<code>httpx</code>	HTTP client for RAG sync to LanceDB

Node.js Packages (auto-installed)

PACKAGE	PURPOSE
<code>inquirer</code>	Interactive prompts (file selection, checkboxes)
<code>chalk</code>	Colored terminal output
<code>ora</code>	Progress spinners for each pipeline stage
<code>cli-table3</code>	Formatted result tables
<code>boxen</code>	Box-styled summary output

03 Installation

Choose between one-click installation (recommended) or manual setup. Both methods install all dependencies automatically.

Option A: One-Click Install RECOMMENDED

Windows - Open PowerShell as Administrator:

PowerShell

```
irm https://unified-platform-projects.s3.us-east-1.amazonaws.com/tools/sow-tool/install-windows.ps1 | iex
```

Linux / macOS - Open terminal:

Bash

```
curl -sL https://unified-platform-projects.s3.us-east-1.amazonaws.com/tools/sow-tool/install-linux.sh | bash
```

What the installer does: Checks Node.js and Python, installs them if missing, downloads the tool package from S3, extracts to `~/sow-analysis-tool`, installs all npm and pip dependencies, then offers to launch immediately.

Option B: Manual Setup

1 Install Prerequisites

Install Node.js 18+ from nodejs.org and Python 3.10+ from python.org

2 Download the Tool

Get the package from S3 or internal repo and extract to your preferred location.

3

Run Setup Script

Navigate to the tool directory and run the setup script for your OS.

Windows

```
cd ~/sow-analysis-tool .\setup.ps1
```

Linux / macOS

```
cd ~/sow-analysis-tool bash setup.sh
```

Directory Structure After Install

```
~/sow-analysis-tool/           # Main tool directory
├── run.js                     # Interactive CLI entry point
├── pipeline.py                # AI analysis pipeline (8 stages)
├── package.json               # Node.js dependencies
├── setup.ps1 / setup.sh      # Setup scripts
├── output/                   # Generated outputs (per project)
│   └── Project_Name/
│       ├── Analysis_*.json
│       ├── WBS_*.xlsx
│       ├── Checklist_*.xlsx
│       ├── Questionnaire_*.xlsx
│       ├── Deployment_Plan_*.xlsx
│       └── Risks_*.json
└── node_modules/             # Auto-installed

~/sow-analyzer/               # AI engine (sibling directory)
├── app/                       # Python modules
│   ├── analyzer.py           # SOW analysis logic
│   ├── wbs.py                # WBS + deployment plan
│   ├── checklist.py           # Checklist + questionnaire
│   ├── parser.py              # PDF/DOCX parser
│   └── llm_provider.py        # Multi-provider AI interface
├── rag/                       # Knowledge base templates
└── *.json                     # WBS patterns, service maps
```

04 Configuration & AI Models

The tool uses AWS Bedrock for AI processing. Authentication is pre-configured with embedded credentials - no AWS account setup needed by the user.

Available AI Models

MODEL	SPEED	QUALITY	BEST FOR
Claude Sonnet 4 DEFAULT	~4-5 min per SOW	Excellent	Production use, complex SOWs, detailed outputs
Claude Haiku 3.5	~2-3 min per SOW	Good	Quick analysis, simple BOQs, iterative drafts

AWS Regions

REGION	LOCATION	LATENCY FROM INDIA
us-east-1	US East (N. Virginia)	~200ms (recommended)
ap-south-1	Asia Pacific (Mumbai)	~20ms
us-west-2	US West (Oregon)	~250ms

No AWS credentials needed! The tool has pre-configured IAM credentials for Bedrock access. You don't need an AWS account, access keys, or profiles. Just select your preferred region and model during the interactive setup.

Supported Input Formats

FORMAT	EXTENSION	TYPE	PARSING METHOD
PDF	.pdf	SOW	markdown (text extraction + OCR fallback)

FORMAT	EXTENSION	TYPE	PARSING METHOD
Word Document	.docx	SOW	markdown (preserves headings/structure)
Excel Workbook	.xlsx	BOQ	openpyxl (row-by-row data extraction)
Legacy Word	.doc	SOW	markdown (binary format support)

05 How It Works — The Pipeline

The tool runs an 8-stage pipeline. Each stage processes sequentially, with real-time progress displayed in your terminal.



Stage Details

STAGE	NAME	WHAT IT DOES	OUTPUT
1	Parse Document	Converts PDF/DOCX/XLSX into structured text. Splits into sections/chunks for AI processing.	Internal (markdown + chunks)
2	Analyze SOW	AI identifies AWS services, architecture patterns, cost implications, compliance needs, and risks.	Analysis_*.json
3	Generate WBS	Creates a hierarchical Work Breakdown Structure with tasks, phases, durations, and dependencies.	WBS_*.xlsx
4	Checklist	Generates an implementation checklist covering all services, configurations, and validation steps.	Checklist_*.xlsx
5	Questionnaire		Questionnaire_*.xlsx

STAGE	NAME	WHAT IT DOES	OUTPUT
		Creates per-service pre-implementation questions to clarify requirements before starting work.	
6	Deployment Plan	Generates a week-by-week deployment timeline with phases, tasks, owners, and milestones.	Deployment_Plan_*.xlsx
7	Risk Analysis	Extracts risks, clarification needs, and potential blockers from the SOW.	Risks_*.json
8	RAG Sync	Pushes document chunks to LanceDB for cross-project knowledge retrieval.	Cloud (silent)

Interactive Workflow (Step-by-Step)

1

Launch the Tool

Run `npm start` from the tool directory. The branded banner appears.

2

Select Your Document

Enter the full path to your SOW (PDF/DOCX) or BOQ (XLSX) file. The tool validates the file exists and shows its size.

3

Choose Outputs

Use arrow keys + space to select which outputs to generate. All are selected by default for SOW files. BOQ files skip checklist/questionnaire.

4

Name Your Project

Enter a project name (used for output filenames and the output subdirectory).

5

Select Region & Model

Choose AWS region and AI model. Sonnet 4 recommended for quality, Haiku 3.5 for speed.

6

Confirm & Run

Review configuration summary, confirm, and watch the pipeline execute with live progress spinners for each stage.

7

View Results

A formatted table shows all generated files with details. Files are saved in `output/Project_Name/`.

06 Output Files — Detailed Guide

Each output is designed for a specific audience and purpose. All Excel files can be opened in Microsoft Excel, Google Sheets, or LibreOffice Calc.

WBS (Work Breakdown Structure) XLSX

Audience: Project Managers, Delivery Leads

Contains: Hierarchical task list with Phase, Task, Subtask, Duration (days), Dependencies, Owner role, Priority, and Notes columns.

Use case: Import into Jira, Monday.com, MS Project, or any project management tool. Provides the foundation for sprint planning and resource allocation.

Implementation Checklist XLSX

Audience: Implementation Engineers, DevOps

Contains: Service-by-service checklist with Category, Item, Status (To Do/Done), Priority, and Notes. Covers configuration, security, networking, monitoring for each AWS service identified.

Use case: Track implementation progress. Ensure nothing is missed during setup. Use as a handoff document between teams.

Pre-Implementation Questionnaire XLSX

Audience: Solutions Architects, Client-Facing Teams

Contains: Per-service questions organized by category (Compute, Networking, Security, Storage, etc.). Each question has Context, Why It Matters, and Default Answer columns.

Use case: Send to the client before kickoff to gather missing information. Identifies gaps in the SOW that need clarification.

Deployment Plan XLSX

Audience: Delivery Managers, Program Managers

Contains: Week-by-week deployment timeline with Phase, Week, Tasks, Owner, Dependencies, Risks, and Milestones columns.

Use case: Present to client during kickoff. Use for executive reporting. Track against actual progress during implementation.

SOW Analysis JSON

Audience: Technical Leads, Architects

Contains: Structured analysis including: identified AWS services (compute, networking, security, storage, database), architecture patterns, compliance requirements, cost estimate ranges, and service interactions.

Use case: Feed into architecture diagrams. Validate against actual AWS service catalog. Use for effort estimation.

Risk Analysis JSON

Audience: Project Managers, Account Managers

Contains: Identified risks with severity, description, mitigation strategy. Plus clarification items (things the SOW doesn't specify clearly).

Use case: Add to project risk register. Discuss with client during discovery. Flag to management for resource planning.

Output naming convention: All files follow the pattern `{Type}_{ProjectName}.{ext}` .
Example: `WBS_Kerala_Eats_E2E.xlsx` , `Risks_Kerala_Eats_E2E.json`

07 **RAG Integration & Knowledge Base**

Every SOW processed is automatically chunked and stored in a vector database (LanceDB) for future retrieval. This enables cross-project learning and pattern matching.

How RAG Sync Works

- 1

Document is Parsed into Chunks

The SOW is split into semantic sections (architecture, requirements, scope, timeline, etc.) during Stage 1.
- 2

Chunks are Embedded

Each chunk is converted to a 768-dimensional vector using the `nomic-embed-text-v1.5` model on the server.
- 3

Stored in LanceDB sow Collection

Vectors + metadata (document name, section type, title) are stored in the shared SOW knowledge base.
- 4

Available for Future Search

Other tools (Unified Platform, SOW Analyzer web) can search this knowledge base using hybrid vector + keyword search with cross-encoder re-ranking.

RAG Architecture

COMPONENT	DETAILS
Vector DB	LanceDB v0.33 on EC2 (i-0d15f876352a8f41d)

COMPONENT	DETAILS
Embedding Model	nomic-ai/nomic-embed-text-v1.5 (768 dimensions)
Re-Ranker	cross-encoder/ms-marco-MiniLM-L-6-v2
Search Type	Hybrid (Vector + BM25 Full-Text) with Reciprocal Rank Fusion
Collection	sow (shared with other platform tools)
API Endpoint	http://impl-internal-products-*.elb.amazonaws.com/ingest

Offline Mode: If the LanceDB server is unreachable (no internet, VPN not connected, server down), the tool continues normally - RAG sync is skipped silently. All local outputs are still generated. You'll see "LanceDB not reachable - skipped (offline mode)" in the output.

Current Knowledge Base Stats

COLLECTION	CHUNKS	PURPOSE
sow	27,000+	SOW document sections from all processed projects
wbs	33	WBS patterns and templates
terraform_docs	2,600+	Terraform documentation for IaC generation
learned_solutions	29	Implementation patterns learned from past projects

08 One-Click Deployment Details

The one-click installer handles everything - from checking prerequisites to launching the tool. Here's exactly what happens under the hood.

Windows Installer Flow

STEP	ACTION	IF MISSING
1	Check Node.js >= 18	Installs via <code>winget install OpenJS.NodeJS.LTS</code>
2	Check Python >= 3.10	Installs via <code>winget install Python.Python.3.12</code>
3	Install Python packages	<code>pip install openpyxl tiktoken boto3 markitdown python-dotenv</code>
4	Download tool from S3	Downloads <code>sow-analysis-tool.zip</code> and extracts to <code>~/sow-analysis-tool</code>
5	Install Node packages	<code>npm install</code> in tool directory

Linux/macOS Installer Flow

STEP	ACTION	IF MISSING
1	Check Node.js >= 18	Installs via NodeSource apt repo (v22 LTS)
2	Check Python 3	Installs via <code>apt-get install python3 python3-pip</code>
3	Install Python packages	<code>pip3 install openpyxl tiktoken boto3 markitdown python-dotenv</code>
4	Download tool from S3	Downloads <code>sow-analysis-tool.tar.gz</code> and extracts

STEP	ACTION	IF MISSING
5	Install Node packages	<code>npm install</code> in tool directory

S3 Package Location

S3 Bucket

```
s3://unified-platform-projects/tools/sow-tool/ |— sow-analysis-tool.zip # Windows package (124 KB) |— sow-analysis-tool.tar.gz # Linux/Mac package (45 KB) |— install-windows.ps1 # Windows installer script |— install-linux.sh # Linux/Mac installer script
```

Public Access: The S3 bucket has a public read policy on the `tools/*` prefix. No AWS credentials are needed to download the installer or package. The URLs work directly in any browser or terminal.

Running After Installation

Launch the tool

```
cd ~/sow-analysis-tool npm start
```

Or if you want to run the pipeline directly without the interactive CLI:

Direct pipeline execution

```
cd ~/sow-analyzer python3 ~/sow-analysis-tool/pipeline.py \ --file "/path/to/your/SOW.pdf" \ --project-name "My_Project" \ --output-dir "./output/My_Project" \ --outputs "analysis,wbs,checklist,questionnaire,deployment_plan,risks" \ --region us-east-1 \ --model "us.anthropic.claude-sonnet-4-20250514-v1:0" \ --SOW
```

09 Troubleshooting

Common issues and their solutions. Most problems are related to prerequisites or network connectivity.

PROBLEM	CAUSE	SOLUTION
"Node.js not found"	Node.js not installed or not in PATH	Install from <code>nodejs.org</code> . On Windows, restart PowerShell after install. Verify with <code>node --version</code>
"Python not found"	Python not installed or aliased differently	Install from <code>python.org</code> . On Linux, try <code>python3</code> instead of <code>python</code> . Check with <code>python3 --version</code>
"Module not found: openpyxl"	Python packages not installed	Run <code>pip install openpyxl tiktoken boto3 markitdown python-dotenv httpx</code>
"ResourceNotFoundException" from Bedrock	Selected AI model not available in chosen region, or model reached end-of-life	Switch to Claude Sonnet 4 model. Use <code>us-east-1</code> region (most models available there)
"AccessDeniedException" from Bedrock	IAM credentials expired or revoked	Contact the platform team to refresh embedded credentials in <code>pipeline.py</code>
Pipeline hangs at "Analyzing SOW"	Large document taking time / slow network	Normal for 20+ page SOWs. Wait up to 5 minutes. If no progress after 10 min, Ctrl+C and retry
"LanceDB not reachable - skipped"	RAG server unavailable	Not an error - tool works fine offline. All local outputs are generated normally. RAG sync

PROBLEM	CAUSE	SOLUTION
	(internet/VPN issue)	will work next time server is reachable
Empty WBS / few tasks generated	Input document too short or poorly formatted	Ensure the SOW has clear sections (scope, requirements, timeline). Try with a more detailed document
"SOW Analyzer not found"	sow-analyzer/ directory missing	The sow-analyzer/ folder must be in the parent directory of sow-analysis-tool/. Re-run the installer
Excel files won't open	File corrupted during generation	Re-run the pipeline. If persists, try opening in Google Sheets instead of Excel
Windows: "execution policy" error	PowerShell blocks unsigned scripts	Run <code>Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser</code> then retry

Need help? If the issue persists, collect the error message and share with the Workmates Platform team. Include: OS version, Node.js version (`node --version`), Python version (`python3 --version`), and the full error output.

10 Frequently Asked Questions

Answers to common questions about the tool's capabilities, limitations, and usage.

How long does it take to process a document?

Typically 3-5 minutes for a 10-30 page SOW using Claude Sonnet 4. Larger documents (50+ pages) may take 8-10 minutes. BOQ files (Excel) are faster since they skip the analysis stage.

Can I process multiple documents at once?

Currently, the tool processes one document per run. Run it multiple times for multiple SOWs. Each run creates a separate output folder named after the project.

What if my SOW is in a language other than English?

Claude understands multiple languages. The tool will attempt to process non-English SOWs, but output quality is best with English documents. The generated outputs will be in English regardless of input language.

Is my document data sent to the cloud?

Yes - the document text is sent to AWS Bedrock (Claude AI) for processing. It is NOT stored by AWS after processing (Bedrock does not retain input/output data). The chunks ARE stored in the RAG knowledge base for future cross-project search.

Can I skip certain outputs?

Yes! During the interactive workflow (Step 3), use the spacebar to deselect any outputs you don't need. Common: generate only WBS + Deployment Plan for quick project planning.

What's the difference between SOW and BOQ mode?

SOW mode (PDF/DOCX): Full analysis - parses document structure, identifies services, generates all 6 outputs.

BOQ mode (XLSX): Simplified - reads line items from Excel, generates WBS and Deployment Plan only (skips analysis, checklist, questionnaire).

Can I customize the output templates?

The AI generates outputs based on prompts in the `sow-analyzer/app/` modules. Advanced users can modify `wbs.py`, `checklist.py`, or `analyzer.py` to change column headers, add fields, or adjust the AI prompts.

Does it work without internet?

No - internet is required for AI processing (AWS Bedrock API calls). The RAG sync step can be skipped (offline mode), but the core AI analysis requires connectivity.

What happens if I run the tool on the same document twice?

The output files are overwritten in the same project folder. RAG sync adds new chunks (does not deduplicate automatically). Use a different project name for each run if you want to compare results.

Can I use my own AWS credentials instead of embedded ones?

Yes - set environment variables `AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, and `AWS_DEFAULT_REGION` before running. Or configure an AWS profile and modify `pipeline.py`.

What's the maximum document size supported?

Tested up to 80 pages / 50,000 words. The tool uses token-aware chunking - very large documents are split into manageable pieces. If a document exceeds Claude's context window, it processes sections independently and merges results.

How accurate are the generated WBS tasks and timelines?

The AI generates a reasonable starting point based on industry patterns and the specific requirements in your SOW. Always review and adjust durations, dependencies, and task assignments before using in production planning.

Ready to Get Started?

Run the one-click installer and process your first SOW in under 5 minutes.

Windows

```
irm ...install-windows.ps1 | iex
```

Linux / macOS

```
curl -sL ...install-linux.sh | bash
```

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